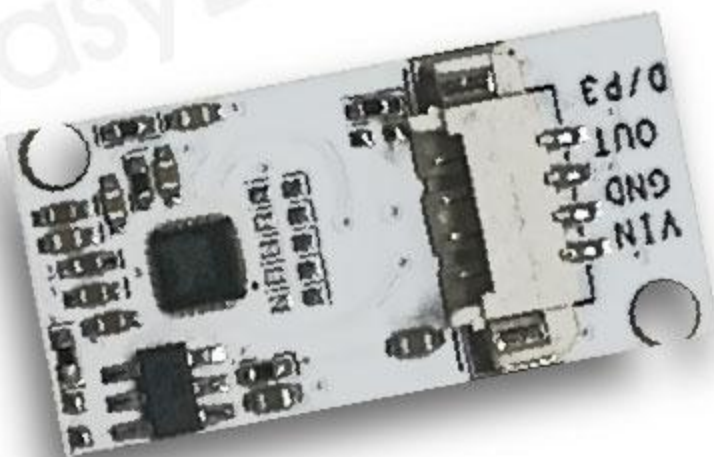


EDX103 Specification

10GHz Low Power Series Modules



Product Features

- 10.5GHz Ultra Low Power Products
- Self-contained logic algorithm for gear differentiation
- Small size, structural design for smart locks

Electrical Parameters

Input Voltage	3.3V±0.1V
Operating Current (convention)	<55uA
Output Voltage	3.3V±0.1V
Output Signal	IO
Power Consumption	<0.5W

Functional Parameters

Motion Sensing Radius ^①	Near gear 0.8m / far gear 1.2m
Hanging height	Front height 1.2m
Delay time	2s

Output Parameters

Center frequency	10.525GHz
3dB Beam Angle	79.7° (XZ plane) 128.7° (YZ plane)

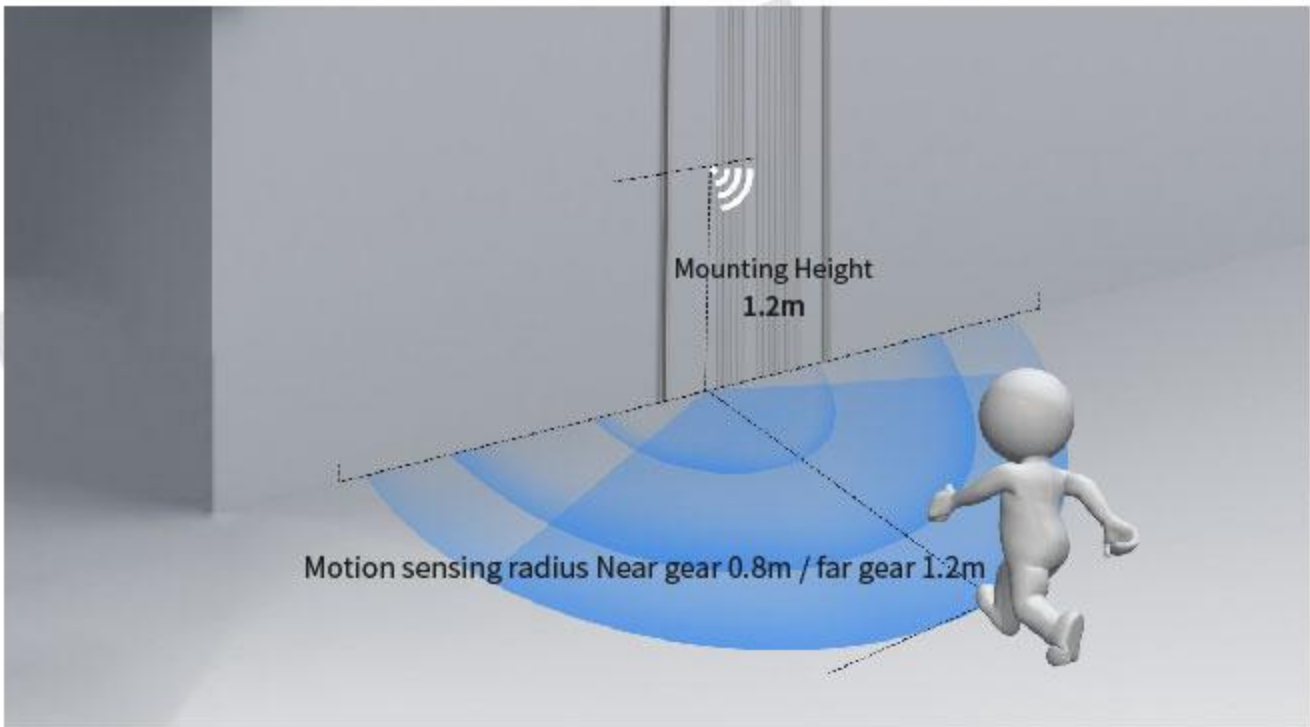
Environment & Lifespan

Operating Temperature	-20~+85°C
Storage Temperature	-20~+105°C

Remarks:

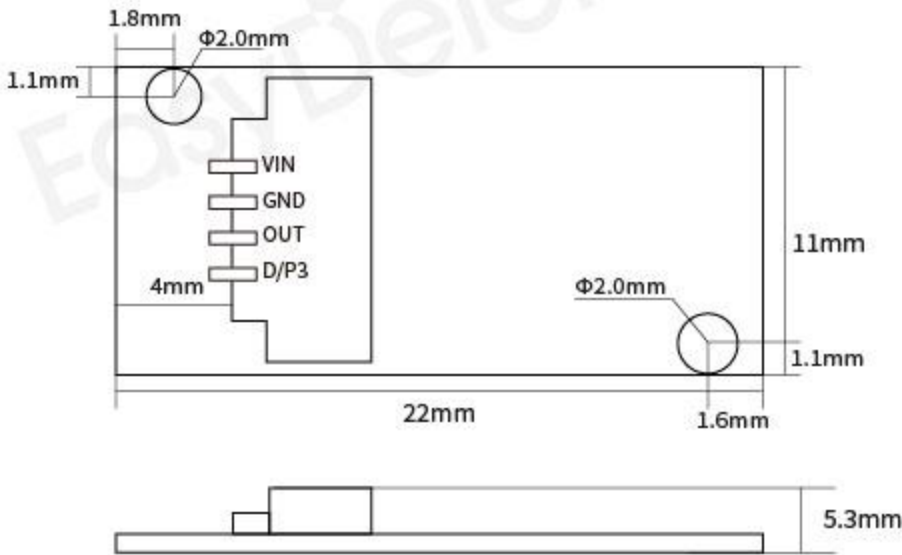
①The test distance range is based on the sensor hanging height of 1.2m, indoor installation environment test, the height of the test person 170cm, weight 65-75kg, walking speed 1m/s, the test person's height 170cm, weight 65-75kg, walking speed 1m/s. Different scenarios may cause changes in the range of installation, subject to the actual test.

Detection Schematic



Dimension Drawing / Pinout

Size unit:mm

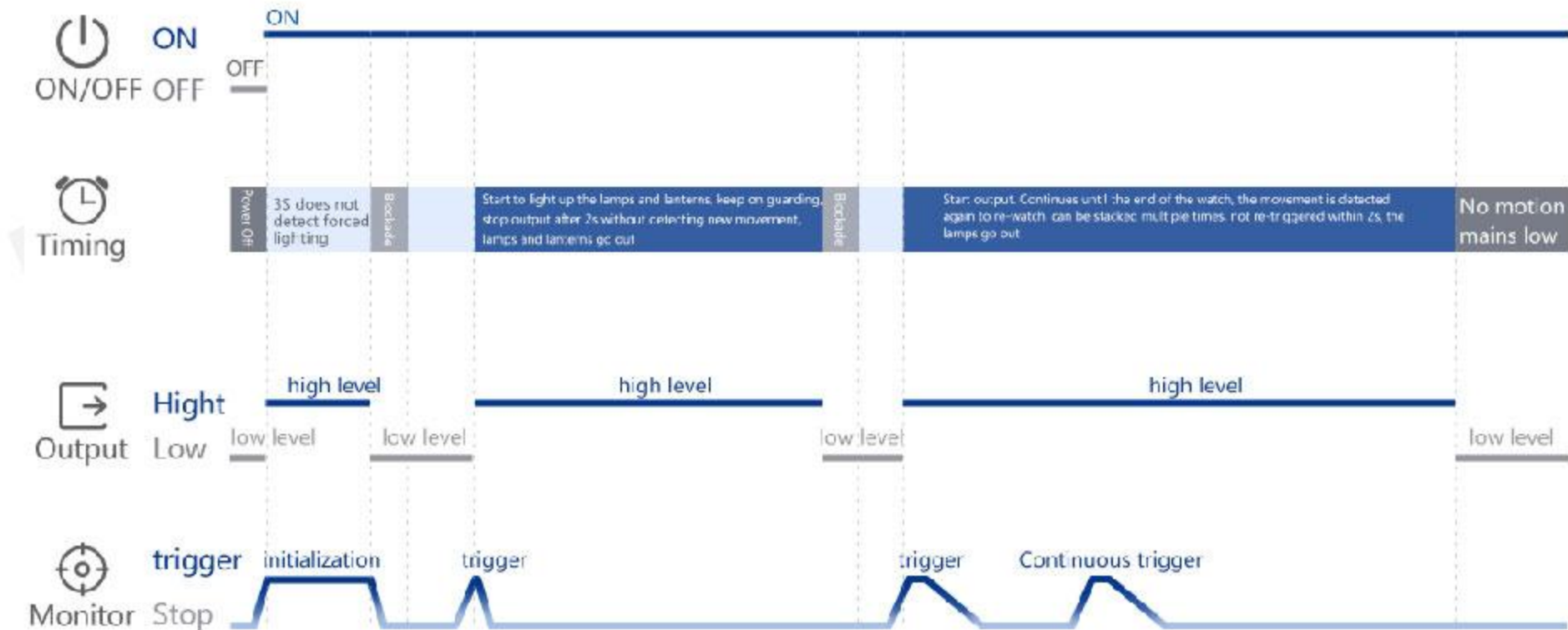


Pin Description

Pin	Description
VIN	Power supply port 3.3V
GND	Groundings
OUT	IO output
D/P3	Gear selection

EDX103 Dimensional tolerance: ±0.2

Timing Diagram



Application Scenarios/Products



corridors



front door



smart lock



peephole

Functional Description



When the sensor detects a moving object, the light automatically lights up and enters the set delay time



After the delay time, if the sensor cannot detect the moving object, the light fixture will turn off

Product Naming Law

ED	Frequency Band	Product Categories	Product Subdivision	Product Number	Delay Time	Serial number
ED	X	1	0	3	N	
EasyDetek	C 5.8GHz	1 Microwave sensor module	0 Ultra-low-power series	0-9,A-Z	Y Has light sensor	
	X 10.5GHz	2. Microwave radar switch	1 Flagship series		N no light sensor	
	Q 24GHz	3 Radar antenna	2 Short-distance series		P programmable	
	V 60GHz	4 MCU	3 Adjustable series			
	W 77GHz	5 Microwave power supply	4 External antenna series			
		6 IC	5 General Series			
		7 Other	6 To be defined			
		8 Networking	7 To be defined			
			8 Basic series			
			9 High altitude series			

Configuration Version Description

【material number】:EDX103-N-02

【hardware】:

【software】:

Historical Revision Record

Version	Time	Descriptive	Note
V1.0	2024-09-24	first edition	

Precautions

1. When installing the product, it is recommended to maintain a distance of 5-12mm between the antenna board and the metal plane, and not to tightly adhere or touch the metal plane.
2. The product has good penetration effect on plastic and wood materials. It is recommended not to install metal, glass, or ceramic in front of the antenna to avoid affecting the actual sensing effect.
3. Please use a power supply with low ripple to avoid sensor interference and false alarms. It is recommended to ensure that the power supply ripple is within 50mV-100V.
4. When multiple radar sensors are applied in the same site, it is recommended that the installation distance of the product be greater than 2m. Installing too close may result in occasional false alarms from individual sensors.
5. The radiation surface of the antenna should avoid being covered by high current circuits to prevent interference with the normal radiation of the antenna, leading to false alarms or changes in the sensing range.
6. When microwave sensors are used in conjunction with wireless communication modules (NB, Bluetooth, WIFI, 2.4G modules), they should be spaced apart. It is recommended to maintain a distance of at least 1m from high-power wireless communication devices such as routers and wireless hotspots during installation.
7. The light sensitivity threshold is the test value under clear weather conditions, no shadows, and diffuse reflection of ambient light.
8. The antenna surface of microwave sensors should avoid facing directly towards the driving power supply, and should also be kept as far away as possible from high-power components such as rectifier bridges, transformers, and switching tubes of the driving power supply to avoid false alarms.
9. EasyDetek Technology is committed to providing customers with high-quality and better experience radar sensors. Product version updates and iterations will not be notified separately.